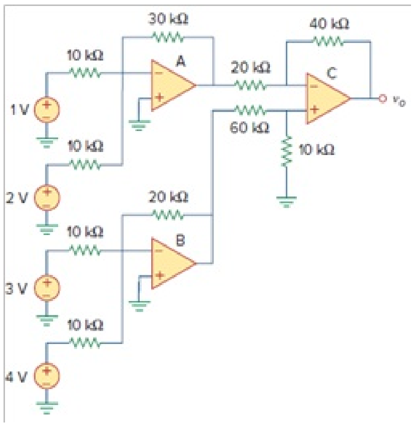
## Problem

Determine *vo* in the op amp circuit of Fig. 5.96.

**Figure. 5.96:**




## Step-by-step solution

**Step 1** of 5

Refer to circuit diagram in Figure 5.96 in the textbook.

Draw the modified circuit as shown in Figure 1.

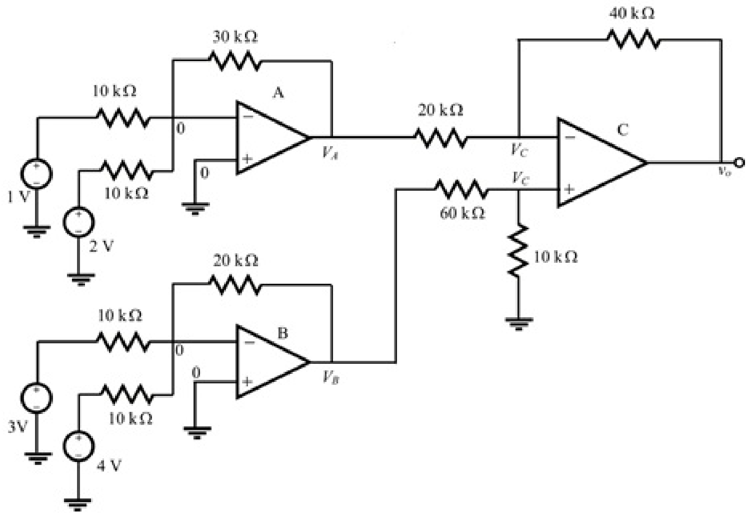



Figure 1

**Step 2** of 5

Apply Kirchhoff's current law at node 0 of op-amp A.

Determine the voltage at $V_A$.

$$\frac{0-1}{10\times10^3}+\frac{0-2}{10\times10^3}+\frac{0-V_A}{30\times10^3}=0$$
$$\frac{V_A}{30\times10^3}=\frac{-1}{10\times10^3}+\frac{-2}{10\times10^3}$$
$$V_A=\left(\frac{-3}{10\times10^3}\right)\left(30\times10^3\right)$$
$$V_A=-9\text{ V}$$

**Step 3** of 5

Apply Kirchhoff's current law at node 0 of op-amp B.

Determine the voltage at $V_B$.

$$\frac{0-3}{10\times10^3}+\frac{0-4}{10\times10^3}+\frac{0-V_B}{20\times10^3}=0$$
$$\frac{-3}{10\times10^3}+\frac{-4}{10\times10^3}=\frac{V_B}{20\times10^3}$$
$$V_B=\left(\frac{-7}{10\times10^3}\right)\left(20\times10^3\right)$$
$$V_B=-14\text{ V}$$

**Step 4** of 5

Apply Kirchhoff's current law at non-inverting terminal of op-amp C.

Determine the voltage at $V_C$.

$$\frac{V_C}{10\times10^3}+\frac{V_C-V_B}{60\times10^3}=0$$
$$\left(\frac{1}{10\times10^3}+\frac{1}{60\times10^3}\right)V_C=\frac{V_B}{60\times10^3}$$

Substitute $-14\text{ V}$ for $V_B$.

$$\left(\frac{6+1}{60\times10^3}\right)V_C=-\frac{14}{60\times10^3}$$

$$\begin{aligned}V_C&=\left(-\frac{14\text{ V}}{60\times10^3}\right)\left(\frac{60\times10^3}{7}\right)\\&=-2\text{ V}\end{aligned}$$

**Step 5** of 5

Determine the voltage at $v_o$.

Apply Kirchhoff's current law at inverting terminal of op-amp C.

$$\frac{V_C - V_A}{20\times10^3} + \frac{V_C - v_o}{40\times10^3} = 0$$

$$V_C\left(\frac{1}{20\times10^3} + \frac{1}{40\times10^3}\right) - \frac{V_A}{20\times10^3} - \frac{v_o}{40\times10^3} = 0$$

$$3V_C - 2V_A - v_o = 0$$

$$v_o = 3V_C - 2V_A$$

Substitute $-2\text{ V}$ for $V_C$, and $-9\text{ V}$ for $V_A$ in equation.

$$\begin{aligned} v_o &= 3(-2) - (2)(-9) \\ &= 12\text{ V} \end{aligned}$$

Therefore, the output voltage $v_o$ is $\boxed{12\text{ V}}$.